OGC METADATA SUMMIT 2026 · FROM METADATA STRATEGY TO PRACTICAL DATA INFRASTRUCTURE

AI-ready metadata in practice

Lessons from building Portolan

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Eurac Research, Bolzano / Bozen · Monday 7 September 2026 · Success stories session

Open Geospatial Consortium

Portolan

CARTO

Who is talking

Twenty-plus years in geospatial: data, science and maps.

- Geospatial engineer at **CARTO**, Seville. Spatial data infrastructures, spatial SQL, cloud-native formats.
- **Portolan contributor**: publisher CLI, catalogs, the OGC Metadata Summit mirror you will see today.
- STAC extensions: `portolan`, `partition`, `osi`, `authentication`; a STAC extension for Apache Iceberg tables.
- Open-source tooling around GeoParquet, DuckDB and raster-in-Parquet.

FIND ME

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THIS TALK

`demo` desk at the coffee breaks, today and tomorrow

`catalog` github.com/cayetanobv/south-tyrol-geodata-portolan-mirror

Everyone says metadata should be AI-ready. We tried to find out what that means.

THE QUESTION

What does "AI-ready metadata" mean *operationally*, for a metadata producer, on a Monday morning?

Not a definition. A set of requirements a machine can check and a publisher can meet.

THE EXPERIMENT · PUBLIC SINCE 2 SEPTEMBER 2026

- a specification, STAC 1.1 at its core
- a validator (`rashid`) and a grader
- a CLI that builds catalogs from files and services
- a registry: a catalog of independently hosted catalogs
- real catalogs, from a city to a planetary archive

What follows is what worked, and what surprised us.

Why we moved: the ambition was right, the experience was painful

WHAT PUBLISHING SPATIAL DATA HAS MEANT

- **For the publisher:** deploy databases, servers, APIs and a portal, then maintain them for years; size for peaks; pay whether the data is used or not; updates are slow.
- **For the user:** find the dataset, understand the service, interpret the metadata, download, reproject, clean, combine. In practice, reachable mainly by GIS specialists.

INSPIRE's ambition — discoverable, interoperable, usable across borders — was right. The infrastructure model underneath it is what became unsustainable.

THREE SHIFTS, NOW CONVERGING

1. **Cloud-native formats changed the economics.** A client reads only the bytes it needs, with its own compute. The publisher stores the data once instead of running a service for every way it might be used.
2. **Agents changed who can use the data.** Given structured metadata, documentation and direct access, an agent finds the authoritative sources, inspects them, runs the analysis and answers with its sources and query. Federation without a central copy.
3. **Sovereignty became an architectural requirement.** Where the data lives, where queries run, which models may touch it, how easily each layer can be swapped.

> The three shifts were already under way. **What was still hard was assembling them into an SDI a real publisher can create, operate and maintain without designing the whole architecture from scratch.** That is why we joined others in building Portolan.

You already know the building blocks. Most carry an OGC stamp.

WHAT A PORTOLAN CATALOG IS MADE OF

STAC 1.1 catalog · collection · item · asset, plus the provider, table, file and web-map-links extensions	OGC Community Standard Oct 2025
every STAC Item is a **GeoJSON** Feature	IETF RFC 7946
Cloud Optimized GeoTIFF for raster	OGC Standard 2023
GeoParquet for vector (Parquet + Simple Features)	OGC Standards Working Group approval pending
PMTiles for map display	community format outside OGC
Zarr · **COPC** (planned, not shipped)	Zarr: OGC Community Standard 2022
HTTP range requests on **object storage** · **CORS**	IETF · W3C

SO WHAT IS LEFT TO SPECIFY?

Every one of those standards *deliberately* leaves publishing choices open: layout, ordering, statistics, documentation, hosting, provenance. Each open choice is a branch every reader must handle.

> **"STAC describes the data.**
> **Portolan defines how it is published."**

A STAC publishing profile, not a new format. Apache-2.0. Spec v0.2.0, 128 requirements, 105 machine-checked.

What Portolan adds, and why it is worth the world's attention

FOUR THINGS THE STANDARDS LEAVE OPEN, MADE MANDATORY

- **A quality floor you can check.** Spatial ordering and bbox columns in every GeoParquet, overviews in every COG, providers and license on every collection, range requests and CORS on the host. `rashid` reads the bytes, not just the JSON.
- **Documentation for two audiences, at every level.** `README.md` for the person deciding whether to trust the data; `AGENTS.md` for the agent that needs the join key and the caveats.
- **No server in the request path.** A folder in storage the publisher controls. Cost is storage and egress; sovereignty is structural; nothing runs after publishing.
- **Federation without a central copy.** A registry of independently hosted catalogs, plain JSON in version control; producer, provider and host named on each.

DOES THE STATIC MODEL SCALE? FIELDS OF THE WORLD

369 TB

published as one Portolan catalog on Source Cooperative

106.6 TB

served in the 28 days to 27 August 2026

No serving layer in front of it.

Data: Fields of the World (fieldsofthe.world), published on Source Cooperative; figure from the Portolan launch post, 2 Sep 2026.

Story 1 · A mirror of this province's open data, published as a spec v0.2.0 catalog in five commands

```
# 1. Pull 69 layers from South Tyrol's WFS (207 available; ~12 m
portolan extract wfs $BZ_WFS ./bz --license CC0-1.0 --workers 4
  --layers "p_bz-Health:*,p_bz-Hydrology:Watercourses,\
            p_bz-TerritorialPlans:UrbanPlan-HazardZonePlan-*,…"

# 2. Validate against the spec
portolan check
# ✗ PTL-PRV-001  no provider carries the 'producer' role  (×69)

# 3. Say who produced it, who hosts it, what it is not;
#    group into 11 themes (a folder move → subcatalogs)
$EDITOR .portolan/metadata.yaml

# 4. Regenerate STAC + README + AGENTS.md, re-check
portolan add --force . && portolan readme
portolan check --fix && rashid check . --schema --data
# 0 error(s), 0 warning(s) across 81 files

# 5. Push to a public bucket (Milan region). Nothing else runs.
portolan push gs://south-tyrol-geodata-portolan-mirror/
rashid check . --live --live-base-url https://storage.googleapis
# 0 error(s): range requests, CORS, sizes all honoured by the ho
```

The service seeded what it could. Titles in three languages, keywords, abstracts, styles from the WMS. 69 layers, 58 MB, converted to GeoParquet, spatially ordered, bbox column added; PMTiles and framed thumbnails alongside.

The validator failed on the machine contract, not on the prose: no producer, no license from the service. The producer's knowledge went into YAML at every level (root, 11 themes, 69 collections); from it came `README.md`, `collection.json`, `AGENTS.md`.

Two independent validators, same answer: conforms to v0.2.0. 128 requirements checked, including the hosting contract once pushed: range requests, CORS, byte sizes. The remaining notes are publisher judgment (no upstream STAC to link to). No server was started. The result is a folder in a bucket.

Data: Provincia Autonoma di Bolzano – Alto Adige / Autonome Provinz Bozen – Südtirol (Ripartizione 28 · ASTAT · Ripartizione 23 Salute), CC0 1.0, via data.civis.bz.it and the provincial WFS. Community mirror extracted 5 September 2026 — not an official publication; the authoritative source is the Province's geoportal.

Story 2 · Then an agent was asked a question no single layer answers

> *"Which hospitals and pharmacies in South Tyrol sit inside high or very-high hydraulic hazard zones of the approved hazard-zone plans, and how many residents live in those municipalities?"*

4 of 17
hospitals inside H3/H4 zones · Bolzano, Bressanone, Brunico, Ortisei

31 of 158
pharmacies inside H3/H4 zones · Merano 3, Bolzano 2, Bressanone 2, Brunico 2 …

108 of 116
municipalities with an approved hydraulic hazard plan · 536,933 residents in the province

STATED BY THE AGENT, BECAUSE THE PUBLISHER WROTE IT DOWN

Points are locations, not service catchments. Hazard classes are planning zones, not event forecasts. Eight municipalities have no approved plan yet. Producer: Autonomous Province of Bolzano – South Tyrol · CC0-1.0 · query attached.

WHAT IT HAD TO WORK WITH

- 69 collections in 11 themes, `AGENTS.md` at every level
- the join key `ISTAT_CODE`, the CRS, the H2–H4 class strings, the caveats
- GeoParquet with bbox columns, read in place with DuckDB

WHAT IT DID NOT HAVE

- a server, an API, credentials, or a person in the loop

A screening, not a risk assessment. The same catalog answers the next question tomorrow.

Data: Provincia Autonoma di Bolzano – Alto Adige — Comuni · Popolazione ufficiale · Piani delle zone di pericolo: pericoli idraulici · Ospedali · Farmacie — CC0 1.0, data.civis.bz.it. Run on a community mirror; a screening, not an official assessment.

What the agent actually did with the metadata

1

Open `catalog.json`

follow the explicit child links: 11 theme catalogs, 69 collections; the linked graph is the publication boundary

2

Read `AGENTS.md`

join key `ISTAT_CODE`, CRS, the H2–H4 class strings, what the data is not for

3

Inspect each collection

picks five of the 69: one GeoParquet asset each, with provider, license, extent, row count

4

Plan a DuckDB query

the model plans and explains; DuckDB computes, reading only the row groups it needs

5

Report

method, exact URLs and query, producer, license, limitations

Every step consumed something the publisher wrote down. Nothing was guessed. That is the operational meaning of AI-ready metadata.

DEMO DESK · COFFEE BREAKS TODAY AND TOMORROW

Both stories live: publish a catalog from *your* WFS, then ask the agent *your* question. And `rashid check --live` against the hosted catalog, to see the hosting contract checked too.

Lesson 1 of 3 · Discovery is the solved half. Evaluability is the gap.

WHAT A GOOD STAC CATALOG ALREADY GIVES YOU

Found. Crawled. Opened.

WHAT PEOPLE AND AGENTS STILL LACK

Is this dataset *for* my question? Where is it weak? How wrong can it be?

ONE SENTENCE FROM A REAL CATALOG (FIELDS OF THE WORLD)

A field polygon "is a *remote-sensing field unit*, **not** a cadastral/legal parcel. This is not a land-tenure product."

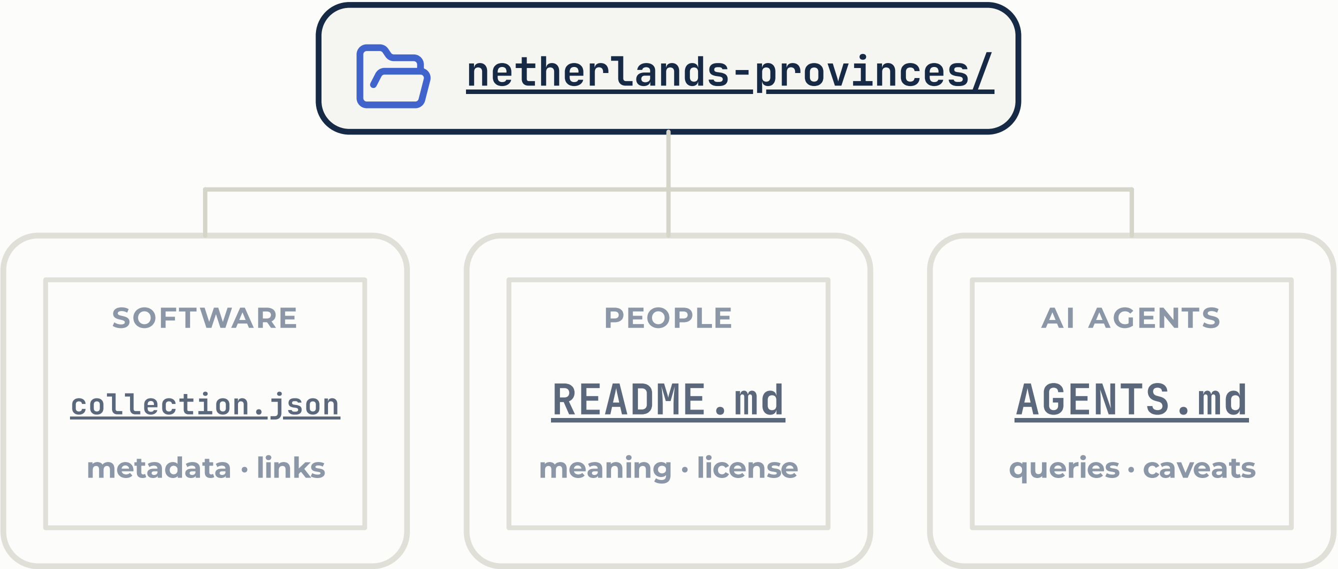
No accuracy figure prevents that misuse. One sentence of scope does.

Source: Fields of the World, global field-boundary dataset documentation (fieldsofthe.world, CC BY 4.0), as quoted in the Portolan documentation best practices.

AI-ready = documented limitations, quantified uncertainty, named biases — **as data where possible** (a confidence column, a quality mask), not buried in prose. Expertise that otherwise lives only in the heads of domain experts.

Lesson 2 of 3 · Two audiences, two files, one subject

ONE COLLECTION · THREE ENTRANCES



THE SAME PUBLISHED ASSETS

GeoParquet · PMTiles · styles · thumbnail · original source

`README.md` **helps a person decide whether to trust the data.** `AGENTS.md` **helps an agent that already committed to using it get the query right the first time. Both generated from one source.**

Lesson 3 of 3 · Be strict, and let a machine hold the line

WHY SO PRESCRIPTIVE

An agent can only rely on what is *required*. Every MAY in a specification is a branch every consumer has to write.

- GeoParquet without bbox columns and spatial ordering: spatial queries read the whole file
- a bucket without CORS: unusable from a browser
- an empty `providers` list: the prose looks fine, every machine fails

WHAT THE MACHINE HOLDS · SPEC v0.2.0

128
normative requirements

105
checked deterministically by `rashid`

23
stay with the publisher's judgment

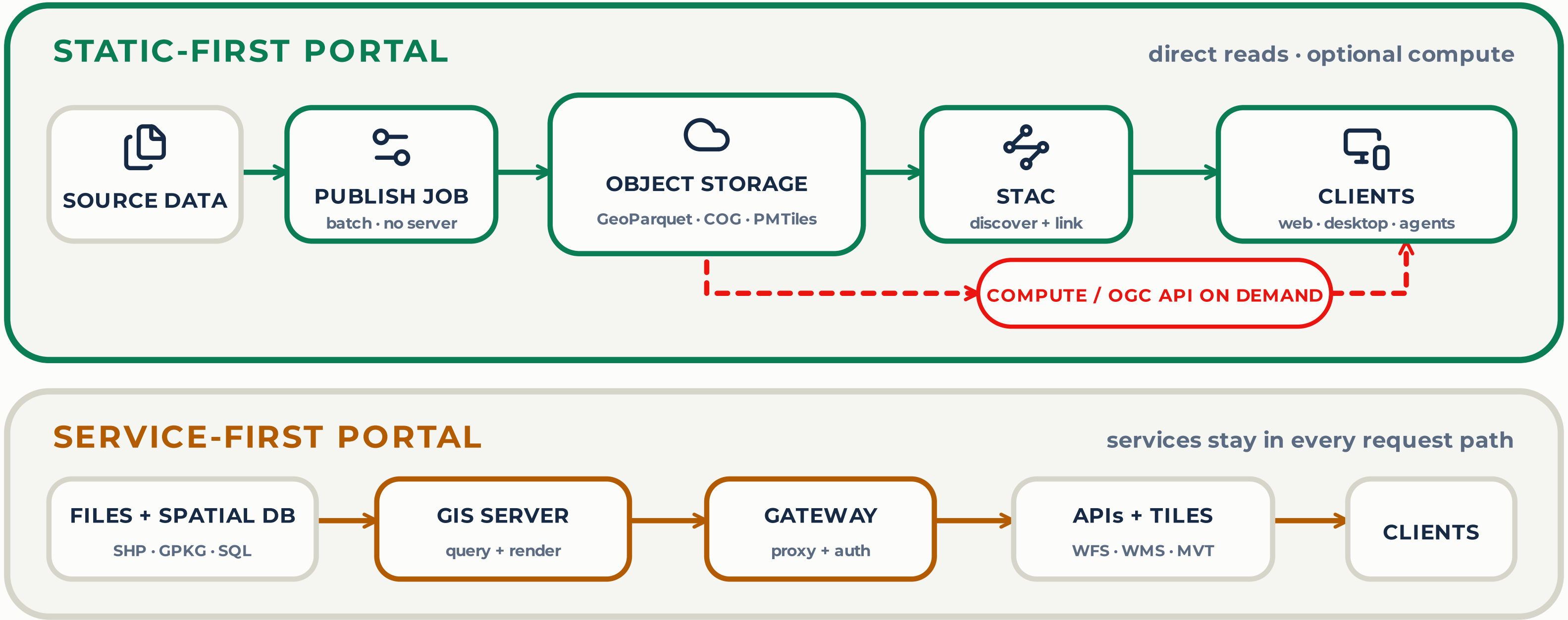
MUST → failure, SHOULD → warning, JSON out. A city's catalog passes the same validator a planetary archive passes.

Conformance is a floor. "You can have perfectly FAIR, utterly useless data." Above pass/fail, a grader scores usability from A+ to F, and weights the machine contract equally with what a browser shows.

A metadata quality policy only sticks when a machine can check it.

Less always-on infrastructure. The work remains.

A NEW PORTAL CAN MAKE STATIC DELIVERY THE DEFAULT



THE QUERY PATH SHRINKS. THE OPERATING WORK REMAINS.

PUBLISH

conversion · metadata · validation

OPERATE

storage · CDN · IAM · DNS · monitor

GOVERN

quality · updates · accountable people

Trust and provenance, honestly unfinished

WHAT EXISTS TODAY

- every collection names producer, licensor, processor and host (STAC provider extension)
- official vs mirror is *derived* from providers; a mirror must name its producer and link back to the original
- the `source` role separates provenance from hosting
- the registry: a catalog of ~20 independently hosted catalogs, plain JSON in version control. "If the registry disappeared tomorrow, every catalog would keep working."

OPEN QUESTIONS · FOR THE TESTBED ON TRUSTED DATA AND SYSTEMS

- How should authority be signalled cryptographically, not just declaratively?
- What does trust mean when the consumer is an agent acting unattended?
- How do mirrors stay verifiably faithful to their source?

Public since 2 September 2026. Early adopters wanted; breaking changes still expected. First goal: ~100 reference catalogs, then a stable v1.0 CLI.

portolan-sdi.org · #portolan on the Cloud-Native Geospatial Forum Slack · community call Fridays 10:00 CET

Questions? Everything is open and one pull request away.

PROJECT LINKS

CLICK ANY CARD

WEBSITE
portolan-sdi.org

ALL PROJECTS
github.com/portolan-sdi

SPECIFICATION
portolan-spec

PUBLISHER CLI
portolan-cli

VALIDATOR
rashid

AGENT SKILLS
portolan-skills

PUBLIC REGISTRY
portolan-registry

CATALOG BROWSER
browser.portolan-sdi.org

THIS DECK

cayetanobv.github.io
gc-metadata-summit-bolzano/#/1

SCAN · CLICK · SHARE

Your ISO 19115 / INSPIRE records are inputs, not casualties

ISO IN

`portolan extract wfs` reads the ISO 19139 record over CSW and seeds title, abstract, keywords, license, contact, lineage, dates.

ISO ALONGSIDE

The record travels with the data as an asset with `roles: ["metadata","iso-19115"]` and a typed `describedby` link to the catalogue that owns it.

GEODCAT-AP OUT

Generated from the same source as a sibling representation, so open-data portals can harvest it. Roadmap, not shipped.

The same pattern swisstopo and the CEOS STAC best practices already follow. There is still no ISO 19115 STAC extension — an open question for this room.

Backup · How an ISO record rides in a Portolan collection

```json
"assets": {
  "metadata": {
    "href": "./iso19115.xml",
    "type": "application/vnd.iso.19139+xml",
    "title": "ISO 19115 record, BRK Bestuurlijke Gebieden (Nationaal Georegister)",
    "roles": ["metadata", "iso-19115"],
    "file:size": 40942, "file:checksum": "1220…"
  }
},
"links": [
  {"rel": "describedby", "href": "./README.md", "type": "text/markdown"},
  {"rel": "describedby",
   "href": "https://www.nationaalgeoregister.nl/geonetwork/srv/api/records/208bc283-…/formatters/xml",
   "type": "application/vnd.iso.19139+xml", "title": "ISO 19115 record — authoritative"},
  {"rel": "via", "href": "https://www.pdok.nl/…", "type": "text/html", "title": "Original source (PDOK)"}
]
```

Asset = pinned copy with checksum and mirror date. Link = live authoritative record. If they disagree, the link wins and the asset tells you how old the copy is.

Lesson 4 · STAC describes the data. Portolan defines how it is published.

STAC 1.1.0 BASELINE + PORTOLAN v0.2.0 PROFILE

DESCRIBES + LINKS

`Catalog → Collection → Item → Asset`

same STAC objects →

PORTOLAN

v0.2.0 · pre-1.0

PROFILES THE PUBLICATION

opinionated · versioned · checkable

THE PROFILE MAKES SIX PUBLISHING CHOICES EXPLICIT

- STRUCTURE + LINKS
- FORMATS + STATISTICS
- ACCESS + HOSTING
- LICENSE + PROVENANCE
- README + AGENTS
- VISUALIZATION

`schemas.portolan-sdi.org/portolan/v0.2.0/schema.json`

An agent can only rely on what is *required*. Every MAY in a specification is a case every consumer must handle defensively.

Lesson 6 · Sometimes "no API" is the honest answer

A PORTOLAN CATALOG IS

Files on any S3-compatible storage. `catalog.json` , `collection.json` , GeoParquet, COG, PMTiles, two Markdown files.

No server to operate. No STAC API. The metadata is the interface.

HOW IT IS READ

An agent reads the catalog, then queries the GeoParquet in place with DuckDB. No credentials. QGIS, Pandas and BigQuery read the same files.

WHAT THAT BUYS

- cost drops to storage and egress
- nothing runs after publishing
- sovereignty is structural: the publisher picks the storage, including national providers
- if Portolan disappeared tomorrow, every file still works